

Project 1 — Build a Narrative Base Model

Train a decoder-only language model from scratch that can continue an incomplete story with recognisably narrative language.

Due: 21 March 2027 · **Weight:** 20%

Page summary

Train a decoder-only language model from scratch that can continue an incomplete story with recognisably narrative language. Given a story prefix of variable length, the model should produce a continuation that is locally coherent, reasonably fluent, connected to the supplied context, and capable of ending at a sensible point.

This is a narrative-domain base model rather than a claim to broad, general-purpose language ability. It becomes the foundation for Project 2's post-training work.

Key facts

Item	Value
Weight	20%
Marks available	20

Item	Value
Training stage	Pre-training from scratch
Work mode	Individual
Model size	32M design target; 33.6M absolute maximum
Compute budget	24 T4-equivalent GPU-hours and 4.0×10^{17} training FLOPs
Main report	15 CVPR pages, excluding references and appendix
Due	Sunday 21 March 2027, 23:59 AET — end of Week 4

What you must demonstrate

Your submission should show that you can turn permitted story data into a working language-model training system, allocate limited model and compute capacity deliberately, compare at least one meaningful choice, and select a final checkpoint using evidence rather than intuition alone.

Tutor-evaluation prompts are authentic prefixes truncated from complete held-out stories. They cover different prefix lengths and story styles within the announced narrative domain. Their source stories and withheld continuations must not influence training or model development.

Design space

Working defaults are supplied for every choice below; change only what you can evaluate within your time and compute budget.

- parameter allocation between depth and width
- feed-forward and attention dimensions
- vocabulary and tokenization

- context length
- data quantity, quality, and narrative mixture
- optimiser and learning-rate schedule
- batch size and gradient accumulation
- warm-up, weight decay, dropout, and gradient clipping
- training duration and checkpoint selection
- end-of-sequence training
- temperature, top-k, top-p, repetition control, and stopping behaviour

Marking breakdown

Component	Marks
Engineering report	10
Submitted model	10
Total	20

Engineering report — 10 marks

1. Pipeline correctness and reproducibility — 2 marks. A correct document-level split, a clearly described preprocessing and tokenization pipeline, a sound from-scratch training setup, and commands that reproduce training, evaluation, and sampling. Marks are reduced when steps are missing, the described setup does not match the repository, leakage is possible, or the system cannot be followed from the evidence provided.

2. Model-data-compute planning — 2 marks. An explicit pre-training target and a reasoned allocation of the model and compute budgets, discussing the relationship between depth/width, context, vocabulary, data, training steps, and expected behaviour. Marks are reduced for listing hyperparameters without explaining their relationship, unrestricted trial and error, or a plan that could not reasonably fit the budget.

3. Experimental rigour — 2 marks. At least one controlled comparison tied to a stated hypothesis, appropriate baselines, consistent evaluation conditions, and useful training evidence (learning curves, stability observations, runtime, processed-token counts). Marks are reduced when several variables change without analysis, comparisons use mismatched conditions, evidence is selectively reported, or conclusions extend beyond the experiment.

4. Evaluation and failure analysis — 3 marks. Correct use of the official reference-token-normalised PPL, BPB or comparable diagnostic loss where informative, representative generated samples with declared decoding settings, and analysis of coherence, repetition, prefix consistency, and stopping failures. Marks are reduced when evaluation depends on one metric, uses only favourable examples, treats the withheld reference continuation as the only valid answer, or does not connect failures to training decisions.

5. Communication and research practice — 1 mark. A concise, readable report that distinguishes measurements, interpretations, and hypotheses; cites datasets, code, and technical claims; labels edited or selected samples honestly; and includes the required AI Assistance Statement.

Submitted model — 10 marks

Submission validity is a technical gate. A valid model receives up to **6⅔ marks** from the shared tutor-evaluation PPL rule and up to **3⅓ marks** from blinded story-continuation review.

6. Tutor-evaluation perplexity — 6⅔ marks. The shared formula is applied to continuation-only reference-token-normalised PPL across all ten tutor prompts: $P \leq 25$ earns all 6⅔ marks; $25 < P < 50$ receives the published exponential decay; $P \geq 50$ earns zero for this criterion.

7. Story continuation quality — 3⅓ marks. Blinded review considers fluency, connection to the given characters, events, and tone, local and short-range consistency, avoidance of empty or repetitive degeneration, and whether the continuation reaches or approaches a reasonable stopping point. Task-specific dimension weights: connection to the supplied prefix **35%**; narrative coherence

and fluency **30%**; avoidance of repetition or degeneration **20%**; stopping and completeness **15%**. Multiple plausible continuations can receive full credit; reproducing the withheld source ending is not required.

Individual work

This project is submitted individually. AI assistance is permitted when declared in the required AI Assistance Statement; responsibility for the submitted work remains with you. See [Policies](#) for the full individual-work, collaboration, and AI-assistance rules.

The spec

The brief poses the problem and leaves room for a response. The spec is the fixed part: what a marker checks. Some lines can be checked mechanically; the rest call for judgement.

- the checkpoint loads with the supplied command and is at or under 33.6M learned parameters
- preprocessing, tokenization, training, evaluation, and sampling can be reproduced from the report's commands
- at least one controlled comparison is tied to a stated hypothesis, with a meaningful baseline
- tutor-evaluation continuation-only PPL is calculated with the published reference tokenizer and formula
- representative and failed generated samples are shown with their decoding settings declared
- the report distinguishes measurements from interpretation and includes the AI Assistance Statement
- training and evaluation stay within 24 T4-equivalent GPU-hours and 4.0×10^{17} training FLOPs
- the ten tutor-evaluation prompts and their withheld continuations never influenced training or model selection

Resources

START HERE

Full Project Brief

Printable copy of the complete task, constraints, evaluation rules, and rubric.

Will be available at 22 February 2027 at 9:00 am AET

Project 1 Starter Repository

Assigned code, default configuration, parameter preflight, budget ledger, and submission manifest.

Will be available at 22 February 2027 at 9:00 am AET

DATA AND MODEL

Narrative Dataset and Data Card

Versioned story corpus with provenance, licence, document-level splits, checksums, and preprocessing notes.

Will be available at 22 February 2027 at 9:00 am AET

Pre-training Colab

Supported route from environment validation to tokenisation, training, checkpointing, evaluation, and sampling.

Will be available at 22 February 2027 at 9:00 am AET

EVALUATE

Public Evaluation Pack

Five development examples, ten tutor prompts, separation validator, and public evaluation utilities.

Will be available at 22 February 2027 at 9:00 am AET

SUBMIT

CVPR Report Template

Shared two-column LaTeX template with the required engineering-report and AI Assistance Statement sections.

Will be available at 22 February 2027 at 9:00 am AET

Submission

Will be available at 1 March 2027 at 12:00 am AET.

Due: 21 March 2027 at 11:59 pm AET

RELATED

[Project 2 – Give the Model a Voice](#)

[Opening lecture](#)

[First review](#)

[Second lecture](#)